

Software Requirements Specification

For

AI-Powered Border Surveillance System with Real-Time Anomaly Detection

Version 1.0

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1. Introduction

This Software Requirements Specification (SRS) describes the complete requirements for the AI-Powered Border Surveillance System with Real-Time Anomaly Detection. The system uses YOLOv8 deep learning, machine learning algorithms, and Microsoft Azure cloud services to provide automated video surveillance, object detection, anomaly detection, and real-time alerting.

1.1 Purpose

The purpose of this SRS document is to:

- Provide comprehensive description of the system's functionality and constraints
- Serve as contractual basis between development team and stakeholders
- Enable developers to design and implement the system correctly
- Provide QA team with testable requirements and acceptance criteria
- Serve as documentation for GTU project evaluation and MS Elevate Program

Intended audience: Developers, project managers, QA testers, system administrators, Prof. Namrata Shroff (Internal Guide), MS Elevate trainers, and GTU evaluation panel.

1.2 Scope

Product Name: AI-Powered Border Surveillance System (ABSS)

The AI-Powered Border Surveillance System is an intelligent, cloud-native surveillance solution for border security monitoring. Built on Microsoft Azure, it leverages computer vision and ML.

The system provides:

- **Automated Object Detection:** Real-time detection of persons, vehicles, suspicious objects using YOLOv8 (75%+ mAP)
- **Anomaly Detection:** Identification of unusual patterns using Isolation Forest + Random Forest
- **Intelligent Alerting:** Email notifications via SendGrid for high-confidence detections (>80%)
- **Cloud Storage:** Azure Blob Storage with lifecycle management (Hot → Cool tier after 30 days)
- **NoSQL Database:** Azure Cosmos DB for detection metadata with date-based partitioning
- **Power BI Dashboard:** Real-time metrics, trends, and interactive filtering
- **Serverless Architecture:** Auto-scaling Azure Functions for processing

- **Enterprise Security:** Azure Key Vault, Managed Identity, encryption at rest/transit

Project Context:

Developed as part of 12-week Microsoft Elevate Program internship (Jan-Apr 2026) at MS Elevate. Demonstrates practical application of 420 hours training across Data Analytics, ML, DL, and Azure Cloud.

1.3 Definitions, Acronyms and Abbreviations

Term/Acronym	Definition/Meaning
ABSS	AI-Powered Border Surveillance System
YOLOv8	You Only Look Once v8 - Real-time object detection algorithm by Ultralytics
CNN	Convolutional Neural Network - Deep learning architecture for images
mAP	Mean Average Precision - Object detection accuracy metric
FPS	Frames Per Second - Video processing speed
Azure	Microsoft Azure - Enterprise cloud computing platform
Blob Storage	Azure Binary Large Object Storage - Cloud object storage
Cosmos DB	Azure Cosmos DB - Globally distributed NoSQL database
Azure Functions	Serverless compute service for event-driven code execution
Power BI	Microsoft Business Intelligence platform for visualization
Anomaly Detection	ML technique for identifying unusual patterns
Isolation Forest	Unsupervised ML algorithm for anomaly detection
Random Forest	Ensemble ML algorithm using decision trees
SendGrid	Cloud email delivery service
Key Vault	Azure service for secure secret storage

1.4 References

- [1] IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications
- [2] Redmon, J., & Farhadi, A. (2018). YOLOv3: An Incremental Improvement. arXiv:1804.02767
- [3] Jocher, G. (2023). Ultralytics YOLOv8. <https://github.com/ultralytics/ultralytics>
- [4] Microsoft Azure Documentation. <https://docs.microsoft.com/azure>
- [5] Liu, F.T., Ting, K.M., & Zhou, Z.H. (2008). Isolation Forest. IEEE ICDM

[6] MS Elevate Program Materials. Edunet Foundation & FICE Education, 2026

1.5 Overview

Section 2 - Overall Description: High-level overview including product perspective, functions, users, environment, and constraints.

Section 3 - Specific Requirements: Detailed functional and non-functional requirements for design and testing.

2. Overall Description

2.1 Product Perspective

Standalone cloud-native application on Microsoft Azure with serverless architecture eliminating on-premises infrastructure.

Architecture Components:

- **Storage Layer:** Azure Blob Storage for video files with tiered storage (Hot/Cool)
- **Processing Layer:** Azure Functions (Python 3.10) for YOLOv8 detection and anomaly detection
- **Data Layer:** Azure Cosmos DB (SQL API) for detection metadata with date partitioning
- **Visualization Layer:** Power BI Service for interactive dashboards
- **Notification Layer:** SendGrid for email alerts
- **Security Layer:** Azure Key Vault for secrets, Managed Identity for auth
- **Monitoring Layer:** Application Insights for telemetry and monitoring

Data Flow: Video Upload → Blob Trigger → Function (Detection) → Cosmos DB + Blob → SendGrid → Power BI

2.2 Product Functions

- **Real-Time Object Detection:** YOLOv8n model fine-tuned on 850 frames, 75%+ mAP, 3 classes
- **Anomaly Detection:** Ensemble ML (Isolation Forest + Random Forest) for spatial/behavioral anomalies
- **Automated Alerts:** SendGrid email with timestamp, confidence, thumbnail for >80% detections
- **Cloud Storage:** Scalable storage with Hot tier, automatic Cool tier after 30 days
- **Interactive Dashboard:** KPIs, charts, timeline, filtering slicers, 15-min refresh
- **Serverless Processing:** Event-driven auto-scaling, handles 10+ concurrent uploads
- **Enterprise Security:** Encryption at rest/transit (TLS 1.2), RBAC
- **Performance Monitoring:** Real-time tracking, error logs, custom telemetry

2.3 User Characteristics

1. Security Personnel

- Role: Monitor dashboard for alerts

- Expertise: Basic computer literacy
- Usage: Daily, 24/7 monitoring
- Training: Minimal (intuitive dashboard)

2. System Administrators

- Role: Deploy, configure, maintain Azure infrastructure
- Expertise: Advanced (Azure proficiency, basic Python)
- Usage: Weekly maintenance, on-demand troubleshooting
- Training: Azure fundamentals (AZ-900 recommended)

3. Data Analysts

- Role: Analyze historical data, generate reports
- Expertise: Intermediate (Power BI proficiency)
- Usage: Weekly/monthly analysis
- Training: Power BI basics (2-4 hours)

2.4 Operating Environment

- **Cloud Platform:**Microsoft Azure (Central India region)
- **Runtime:**Azure Functions with Python 3.10 runtime
- **Client:**Web browser (Chrome 90+, Firefox 88+, Edge 90+)
- **Video Input:**MP4, AVI, MOV; 720p min, 1080p recommended
- **Network:**Internet connectivity (10 Mbps min for uploads)

2.5 Design and Implementation Constraints

- **Budget:**Must operate within Azure for Students credits (₹15,600)
- **Languages:**Python 3.10 for ML/DL, JavaScript for dashboard
- **ML Framework:**YOLOv8 (Ultralytics) mandatory
- **Cloud:**Microsoft Azure ecosystem only
- **Serverless Limits:**5-minute timeout, 1.5GB memory limit
- **Security:**All data encrypted at rest and in transit

2.6 Assumptions and Dependencies

Assumptions:

- Cameras provide consistent 720p+ video quality
- Network connectivity stable for uploads
- Azure maintains 99.9% uptime SLA
- Users have basic computer literacy

Dependencies:

- Microsoft Azure platform availability
- YOLOv8 Ultralytics library (v8.0+)
- Azure SDK for Python (azure-storage-blob, azure-cosmos)
- SendGrid email API
- Power BI Service for dashboard hosting

3. Specific Requirements

3.1 Functionality

3.1.1 Video Upload and Storage

- **FR-1.1:**System shall accept MP4, AVI, MOV formats
- **FR-1.2:**System shall store videos in Azure Blob Storage Hot tier
- **FR-1.3:**System shall move videos to Cool tier after 30 days
- **FR-1.4:**System shall support files up to 500MB

3.1.2 Object Detection Processing

- **FR-2.1:**System shall auto-trigger YOLOv8 on video upload
- **FR-2.2:**System shall extract frames at 1 FPS
- **FR-2.3:**System shall detect person, vehicle, suspicious_object
- **FR-2.4:**System shall assign confidence scores (0-100%)
- **FR-2.5:**System shall save processed frames with bounding boxes

3.1.3 Anomaly Detection

- **FR-3.1:**System shall use Isolation Forest + Random Forest ensemble
- **FR-3.2:**System shall flag unusual spatial locations
- **FR-3.3:**System shall mark >80% confidence as priority

3.1.4 Data Storage

- **FR-4.1:**System shall store results in Cosmos DB: video_id, timestamp, class, confidence, bbox, anomaly_flag
- **FR-4.2:**System shall partition data by date
- **FR-4.3:**System shall maintain upload time, duration, error logs

3.1.5 Alerts and Notifications

- **FR-5.1:**System shall send emails via SendGrid for >80% confidence
- **FR-5.2:**Emails shall include timestamp, class, confidence, thumbnail
- **FR-5.3:**System shall log all alerts in Cosmos DB

3.1.6 Dashboard and Visualization

- **FR-6.1:**System shall provide Power BI dashboard with 15-min refresh
- **FR-6.2:**Dashboard shall display KPIs: Total Detections, High Confidence Alerts, Avg Processing Time
- **FR-6.3:**Dashboard shall show bar chart by object class
- **FR-6.4:**Dashboard shall show timeline visualization
- **FR-6.5:**Dashboard shall provide slicers: date, class, confidence
- **FR-6.6:**Dashboard shall display detailed alert table with drill-down

3.1.7 Security and Access Control

- **FR-7.1:**System shall store secrets in Azure Key Vault
- **FR-7.2:**System shall use Managed Identity for service auth
- **FR-7.3:**System shall encrypt data at rest (SSE)
- **FR-7.4:**System shall use HTTPS/TLS for all data in transit

3.2 Usability

- **UR-1:**Dashboard accessible without training for basic users
- **UR-2:**Dashboard responsive and mobile-friendly (tablet support)
- **UR-3:**Video upload completes in <30 seconds for 100MB files
- **UR-4:**System provides clear error messages for failures
- **UR-5:**Dashboard visualizations load within 3 seconds

3.3 Reliability

- **RR-1:**System availability: 99.5% uptime (Azure SLA dependent)
- **RR-2:**MTBF: 720 hours (30 days)
- **RR-3:**Detection accuracy: Minimum 75% mAP@0.5
- **RR-4:**False positive rate: <20% for high-confidence
- **RR-5:**Automated retry for failed uploads (3 attempts)
- **RR-6:**Cosmos DB automatic backups every 4 hours

3.4 Performance

- **PR-1:**Video processing: 30+ FPS for 720p real-time
- **PR-2:**Average processing: 15-20 seconds per 50MB video
- **PR-3:**Cosmos DB queries: <500ms for dashboard data
- **PR-4:**Alert email delivery: Within 10 seconds
- **PR-5:**Handle 10 concurrent uploads without degradation
- **PR-6:**Blob upload speed: Minimum 5 MB/s
- **PR-7:**Power BI refresh: Complete within 15 minutes

3.5 Supportability

- **SR-1:**System shall integrate Application Insights for monitoring
- **SR-2:**Errors logged with severity levels (Info, Warning, Error, Critical)
- **SR-3:**System shall maintain deployment docs and API references
- **SR-4:**Code version controlled on GitHub with README
- **SR-5:**Configuration manageable via Azure Portal without code changes

3.6 Design Constraints

- **DC-1:**Must use YOLOv8 architecture (project requirement)
- **DC-2:**Must deploy on Azure (educational license)

- **DC-3:**Must use Python 3.10 for Azure Functions compatibility
- **DC-4:**Must follow Azure Blob naming conventions (lowercase, no special chars)
- **DC-5:**Power BI dashboard must publish to Power BI Service

3.7 Purchased Components

The system uses Azure for Students subscription (free ₹15,600 credits). All components are cloud services with pay-as-you-go pricing:

- Azure Blob Storage (LRS, Standard tier)
- Azure Functions (Consumption plan)
- Azure Cosmos DB (provisioned throughput, 400 RU/s)
- SendGrid email API (free tier for development)
- Power BI Service (free tier)

3.8 Interfaces

3.8.1 User Interfaces

- **UI-1:**Azure Portal for video upload (drag-and-drop)
- **UI-2:**Power BI web interface (1920x1080 optimal)
- **UI-3:**Color scheme: Blue (#2E75B6) primary, Red (#C00000) alerts

3.8.2 Hardware Interfaces

- **HI-1:**Support for IP cameras, CCTV producing MP4/AVI
- **HI-2:**No direct hardware integration - cloud-based

3.8.3 Software Interfaces

- **SI-1:**Azure Blob Storage API for upload/download
- **SI-2:**Azure Functions runtime for serverless execution
- **SI-3:**Azure Cosmos DB SQL API for NoSQL operations
- **SI-4:**SendGrid Email API v3 for notifications
- **SI-5:**Power BI REST API for dashboard refresh
- **SI-6:**Application Insights SDK for telemetry

3.8.4 Communications Interfaces

- **CI-1:**HTTPS protocol for all Azure communications
- **CI-2:**TLS 1.2+ encryption for data in transit
- **CI-3:**SMTP over TLS for email (SendGrid port 587)
- **CI-4:**JSON format for all API request/response

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